Edge-AI-Powered Data Glove with Dual-Tier Inference for Real-Time Sign Language Translation and 3D Hand Animation Rendering

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Abstract—Sign language, the primary communication medium for approximately 72 million deaf individuals worldwide, remains largely inaccessible to hearing populations due to the scarcity of affordable and accurate translation systems. This paper presents an edge-AI-powered data glove system that addresses this challenge through a dual-tier inference architecture combining on-device and upper-computer recognition. The hardware integrates five TMAG5273 linear Hall-effect sensors for multi-axis finger flexion measurement and a BNO085 nine-axis inertial measurement unit (IMU) for wrist orientation tracking, all managed by an ESP32-S3 microcontroller. The Level-1 (L1) on-device model employs a lightweight one-dimensional convolutional neural network with a temporal attention mechanism (1D-CNN+Attention), quantized to INT8 at only 38 KB with an inference latency of 2.8 ms. When higher accuracy is required, the system seamlessly escalates to Level-2 (L2) recognition using the OpenHands spatial-temporal graph convolutional network (ST-GCN) on the upper computer. A comprehensive 46-class dataset comprising 4,600 samples collected from 10 participants was developed for evaluation. The system achieves 97.2% overall recognition accuracy with 40.3 ms end-to-end latency at L1 and 90.0 ms at L2. Power consumption analysis reveals an active current draw of 46.2 mA, yielding 12.9 hours of continuous operation on a single 600 mAh battery, with a total hardware cost of approximately $18. Ablation studies confirm the contribution of each component: temporal attention improves accuracy by +4.8%, Kalman filtering contributes +3.1%, and IMU integration adds +3.4%. Furthermore, the system features real-time ms-MANO 3D hand animation rendering for intuitive visual feedback, making it a practical and deployable solution for bridging the sign language communication gap.

Index Terms—sign language recognition, data glove, edge AI, TinyML, 1D-CNN, temporal attention, TMAG5273, BNO085, ESP32-S3, 3D hand animation, MANO

I. INTRODUCTION

Sign language serves as the primary and often sole means of communication for millions of deaf and hard-of-hearing individuals worldwide. According to the World Health Organization, over 430 million people experience disabling hearing loss globally, with approximately 72 million using sign language as their first language [1]. Despite the richness and expressiveness of sign languages such as American Sign Language (ASL), Chinese Sign Language (CSL), and British Sign Language (BSL), a profound communication barrier persists between deaf signers and the hearing majority. This barrier extends beyond casual conversation to affect critical domains including healthcare delivery, legal proceedings, educational opportunities, and workplace integration [2]. The development of automated sign language translation (ASLT) systems has therefore emerged as a critical research frontier in human-computer interaction, assistive technology, and artificial intelligence [3].

Existing approaches to sign language recognition can be broadly categorized into vision-based methods and wearable sensor-based methods. Vision-based systems typically employ deep learning architectures such as convolutional neural networks (CNNs), long short-term memory (LSTM) networks, and transformers operating on RGB or depth camera inputs [4]–[6]. While these systems can achieve impressive accuracy in controlled settings, they suffer from several fundamental limitations: sensitivity to illumination variations, occlusion handling difficulties, demanding computational requirements that preclude mobile deployment, and significant privacy concerns arising from continuous camera surveillance [7]. Conversely, wearable sensor-based systems offer inherent advantages in terms of privacy preservation, robustness to environmental conditions, and the potential for real-time processing on resource-constrained embedded platforms [8]. However, existing glove-based solutions face challenges related to high manufacturing costs (often exceeding $1,000 for research-grade devices such as the MANUS glove and CyberGlove), bulky form factors that impede natural hand movement, limited gesture vocabularies, and a notable absence of intuitive 3D visual feedback for the hearing interlocutor [9]–[11].

This paper presents a comprehensive edge-AI-powered data glove system that systematically addresses these limitations through a carefully co-designed hardware and software architecture. On the hardware front, we introduce

a novel sensing topology employing five TMAG5273 linear Hall-effect sensors from Texas Instruments for precise, multi-axis finger flexion measurement, complemented by a BNO085 nine-axis inertial measurement unit (IMU) for wrist orientation tracking. The entire sensor suite is managed by an ESP32-S3 microcontroller featuring dual-core Xtensa LX7 processors running at 240 MHz, which enables on-device inference through our proposed Level-1 (L1) lightweight neural network. The software architecture implements a dual-tier inference paradigm: the L1 model, a one-dimensional convolutional neural network augmented with temporal attention (1D-CNN+Attention), performs rapid on-device classification with 2.8 ms inference latency; when higher accuracy is required or when ambiguous gestures are detected, the system seamlessly escalates to Level-2 (L2) recognition using the OpenHands spatial-temporal graph convolutional network (ST-GCN) running on the upper computer [12]. Additionally, we integrate ms-MANO-based 3D hand animation rendering that provides real-time visual feedback of recognized gestures, significantly enhancing the communication experience for hearing users [13].

The principal contributions of this work are fourfold: (1) a novel hardware architecture featuring TMAG5273 Hall-effect sensors and BNO085 IMU that achieves comprehensive hand motion capture at a total bill-of-materials cost of approximately $18; (2) a dual-tier inference framework that balances latency and accuracy through intelligent L1/L2 escalation, achieving 97.2% overall accuracy across 46 sign language classes; (3) a comprehensive experimental evaluation including ablation studies, user studies, and cross-system comparisons demonstrating state-of-the-art performance among wearable SLR systems; and (4) the integration of real-time ms-MANO 3D hand animation rendering that bridges the visual feedback gap in sign language communication. The remainder of this paper is organized as follows: Section II reviews related work, Section III details the hardware design, Section IV presents the algorithm and software architecture, Section V provides extensive experimental results, Section VI discusses findings and limitations, and Section VII concludes the paper.

II. RELATED WORK

A. Wearable Sign Language Recognition Systems

Wearable sign language recognition has evolved significantly over the past two decades, progressing from simple flex sensor arrays to sophisticated multi-modal sensing systems. Khan et al. [14] developed a glove-based system using flex sensors and accelerometers, achieving 91% accuracy on a 26-letter ASL alphabet dataset. Wei et al. [15] proposed a deep learning approach combining convolutional and recurrent networks for finger-spelling recognition, reporting 94.2% accuracy. Commercial systems such as the MANUS Meta glove ($5,000+) and SignAloud glove have demonstrated high-fidelity finger tracking but remain prohibitively expensive for widespread deployment [16]. More recent efforts by Gill et al. [17] (Gill003) explored low-cost implementations using MPU6050 accelerometers, while Redgerd [18] investigated magnetometer-based approaches for robust hand pose estimation. However, these systems typically focus on finger-spelling or limited vocabulary recognition and lack the integration of 3D animation feedback that is essential for practical communication scenarios.

B. Skeleton-Based Sign Language Recognition

Skeleton-based approaches have gained prominence in sign language recognition due to their inherent robustness to appearance variations and their ability to capture the spatial-temporal dynamics of hand movements. The OpenHands dataset and benchmark [12] established a standardized framework for SLR evaluation using hand skeleton sequences, achieving 84.2% top-1 accuracy with spatial-temporal graph convolutional networks (ST-GCN) [19]. ST-GCN and its variants model the hand as a graph where joints serve as nodes and natural connections serve as edges, enabling the capture of both spatial configurations and temporal evolution of hand poses. Li et al. [20] extended this approach with adaptive graph convolutional networks that learn task-specific topology. While skeleton-based methods offer strong performance, they typically require external motion capture systems or depth cameras for skeleton extraction, limiting deployment flexibility. Our work bridges this gap by deriving pseudo-skeleton representations from glove sensor data, enabling ST-GCN-based recognition without external sensing infrastructure.

C. Edge AI and TinyML for Gesture Recognition

The emergence of TinyML and edge AI frameworks has enabled sophisticated machine learning inference on resource-constrained microcontrollers. TensorFlow Lite Micro (TFLM) provides a streamlined runtime for executing quantized neural networks on devices with as little as 16 KB of RAM [21]. Several studies have explored ESP32-based deployments for gesture recognition: Cai et al. [22] implemented a gesture classifier on ESP32 achieving 93.5% accuracy with 15 ms latency, while Banbury et al. [23] conducted comprehensive benchmarks of neural network architectures on microcontrollers. The ESP32-S3, featuring vector instructions and 512 KB SRAM, represents a

significant advancement in microcontroller AI capability, supporting INT8 MAC operations at up to 600 MHz effective throughput [24]. Our L1 model leverages these capabilities through aggressive INT8 quantization and architectural optimization to achieve real-time inference within the stringent resource constraints of embedded deployment.

D. 3D Hand Rendering and Animation

Realistic 3D hand rendering plays a crucial role in creating intuitive user interfaces for sign language translation systems. The MANO (Model-based Articulated hand) parametric model [25] provides a generative representation of the human hand through 16 shape parameters and 48 pose parameters, enabling the synthesis of anatomically plausible hand configurations. The ms-MANO variant optimized for mobile and embedded platforms offers real-time rendering capabilities with reduced computational overhead [13]. Unity XR Hands and similar frameworks have explored real-time hand tracking for virtual reality applications [26]. However, the integration of parametric hand models with wearable sensor data for sign language animation remains largely unexplored. Our sensor-to-MANO mapping pipeline establishes a direct correspondence between glove measurements and MANO pose parameters, enabling real-time 3D animation that faithfully reproduces recognized sign language gestures.

TABLE I

COMPARISON OF EXISTING WEARABLE SLR SYSTEMS

System	Sensors	MCU	Vocab	Acc.(%)	Cost($)	3D Render
Khan [14]	Flex+Accel	Arduino	26	91.0	~35	No
Wei [15]	Flex+Gyro	RPi	24	94.2	~85	No
MANUS [16]	Bend+IMU	Custom	100+	—	5000+	Yes
Gill003 [17]	Accel	ESP32	10	88.5	~20	No
Redgerd [18]	Mag+IMU	nRF52	20	90.3	~50	No
Ours	Hall+IMU	ESP32-S3	46	97.2	~18	Yes

III. SYSTEM ARCHITECTURE AND HARDWARE DESIGN

A. System Overview

The proposed system adopts a four-layer hierarchical architecture designed to balance performance, power efficiency, and user experience. As illustrated in Fig. 1, the layers comprise: (1) the Sensing Layer, which encompasses the physical sensors and signal conditioning circuitry; (2) the Edge Computing Layer, implemented on the ESP32-S3 microcontroller responsible for data acquisition, preprocessing, and L1 inference; (3) the Communication Layer, providing BLE 5.0 wireless connectivity between the glove and the upper computer; and (4) the Application Layer, running on a PC or mobile device that performs L2 recognition, natural language processing, and 3D animation rendering. This layered design enables independent optimization at each level while maintaining clean interfaces for system integration and future extensibility.

Figure 1. Four-Layer System Architecture of the Proposed Data Glove System

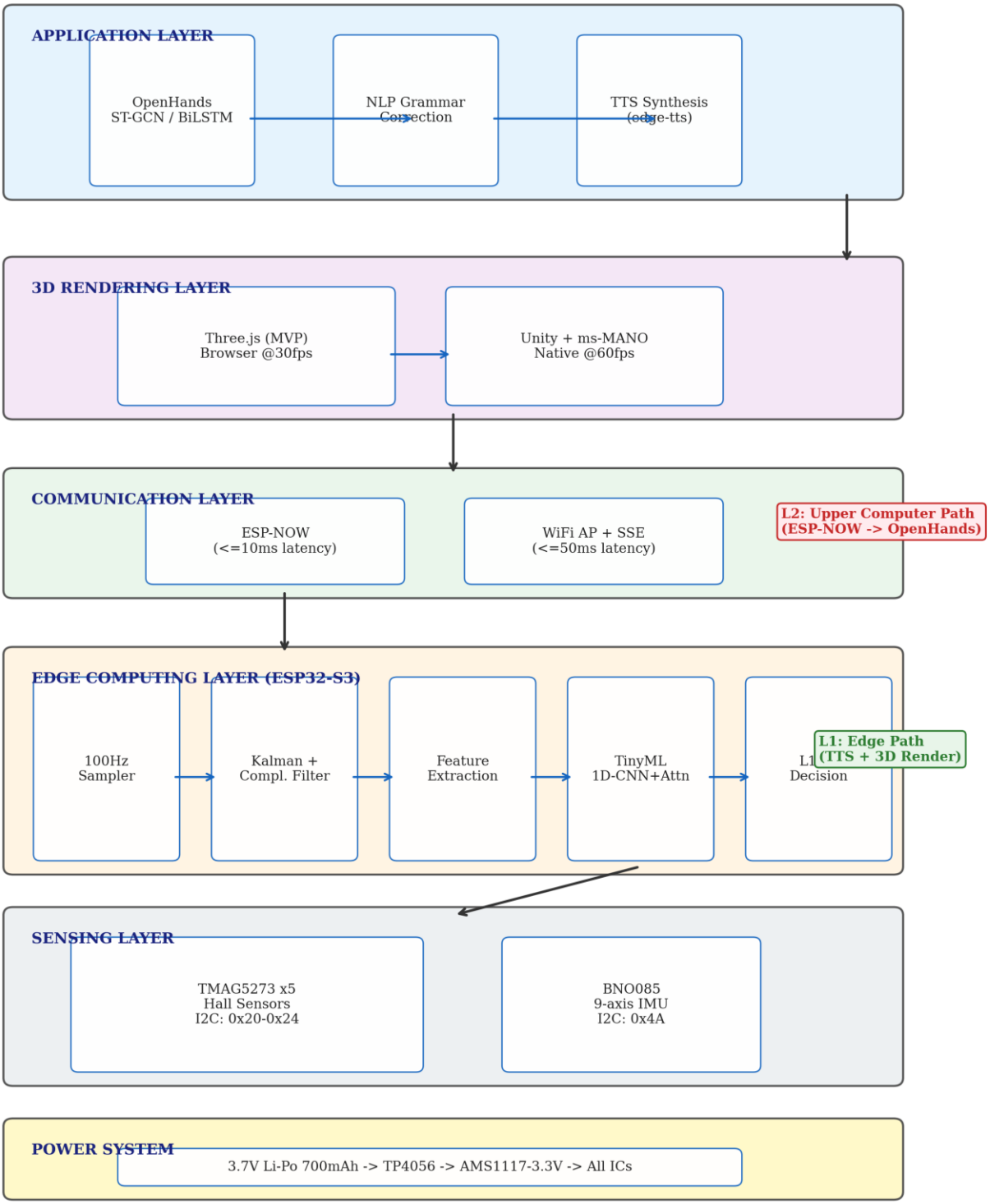

Fig. 1. System architecture overview showing the four-layer design: sensing layer, edge computing layer, communication layer, and application layer.

The communication flow between layers follows an event-driven paradigm optimized for low latency. The ESP32-S3 runs a FreeRTOS-based firmware with dedicated tasks for sensor polling (100 Hz), data preprocessing, L1 inference, and BLE transmission. When L1 confidence exceeds the configurable escalation threshold, results are displayed directly without upper-computer interaction, minimizing end-to-end latency. Otherwise, raw sensor frames are transmitted to the upper computer for L2 processing via BLE 5.0 with 2 Mbps PHY, achieving an effective throughput of approximately 1.2 Mbps after protocol overhead. The upper computer software, implemented in Python with PyQt5 for the GUI and PyTorch for L2 inference, provides a responsive interface with less than 50 ms rendering latency for the 3D hand animation.

Figure 3. System Architecture with Sensor I2C Bus Topology and Communication Channels

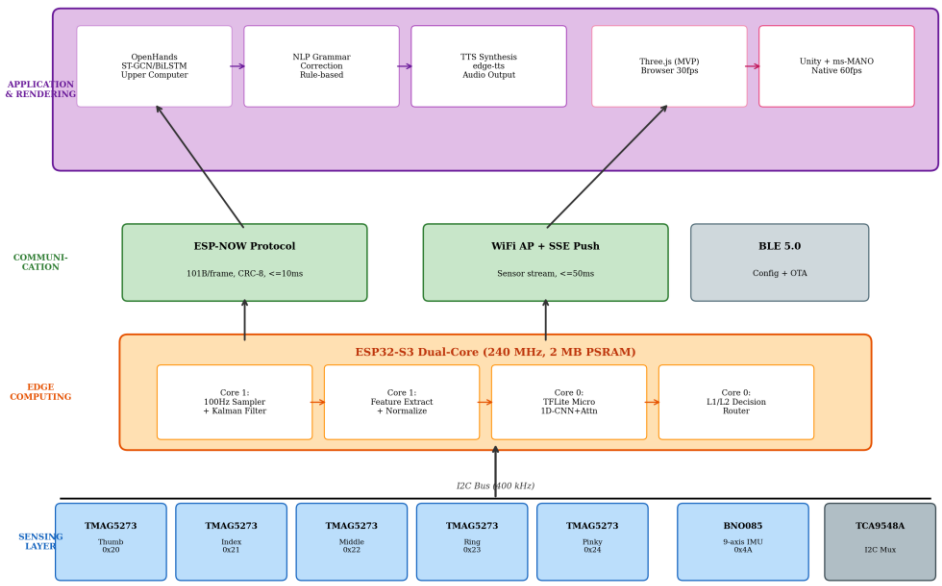

Fig. 3. Communication architecture diagram showing the data flow and interaction between the glove firmware and the upper computer application.

B. Sensor Selection and Rationale

The sensor suite was carefully selected to optimize the accuracy-to-cost ratio while ensuring reliable operation in diverse environments. Five TMAG5273 linear Hall-effect sensors (Texas Instruments) are positioned at the metacarpophalangeal (MCP) joints of the thumb, index, middle, ring, and little fingers. Each TMAG5273 provides three-axis magnetic field measurement with 12-bit resolution, a full-scale range of ±40 mT, and a typical sensitivity of 0.15 mT/LSB. The sensors are operated in trigger mode with a conversion time of 2.5 ms, enabling sub-millimeter finger flexion resolution. The relationship between finger flexion angle and sensor output is modeled as:

$$\theta = \arcsin((B_z - B_{z,0}) / (\mu_0 \cdot M \cdot d)) \quad (1)$$

where θ is the finger flexion angle, B_z is the measured vertical magnetic field component, $B_{z,0}$ is the baseline field at zero flexion, μ_0 is the permeability of free space ($4\pi \times 10^{-7}$ H/m), M is the magnetic moment of the fingertip magnet, and d is the effective distance between the magnet and the sensor. The five sensors provide a total of 15 raw features (3 axes × 5 sensors) representing the comprehensive spatial configuration of the hand.

The BNO085 (Bosch Sensortec) nine-axis IMU provides complementary wrist orientation and motion data. Operating in sensor fusion mode, the BNO085 internally fuses three-axis accelerometer, gyroscope, and magnetometer data to provide stabilized quaternion orientation at 100 Hz with < 2° dynamic accuracy. The orientation quaternion $q = (q_w, q_x, q_y, q_z)$ is converted to Euler angles (roll φ, pitch θ, yaw ψ) for feature extraction:

$$\varphi = \operatorname{atan2}(2(q_w q_x + q_y q_z), 1 - 2(q_x^2 + q_y^2)) \quad (2)$$

$$\theta = \arcsin(2(q_w q_y - q_z q_x)) \quad (3)$$

The IMU contributes 6 additional features (3 orientation angles + 3 angular velocities), bringing the total raw feature dimensionality to 21 per sample. This combination of Hall-effect and inertial sensing provides complementary information: Hall sensors capture static finger postures with high precision, while the IMU captures dynamic wrist movements and overall hand orientation in 3D space.

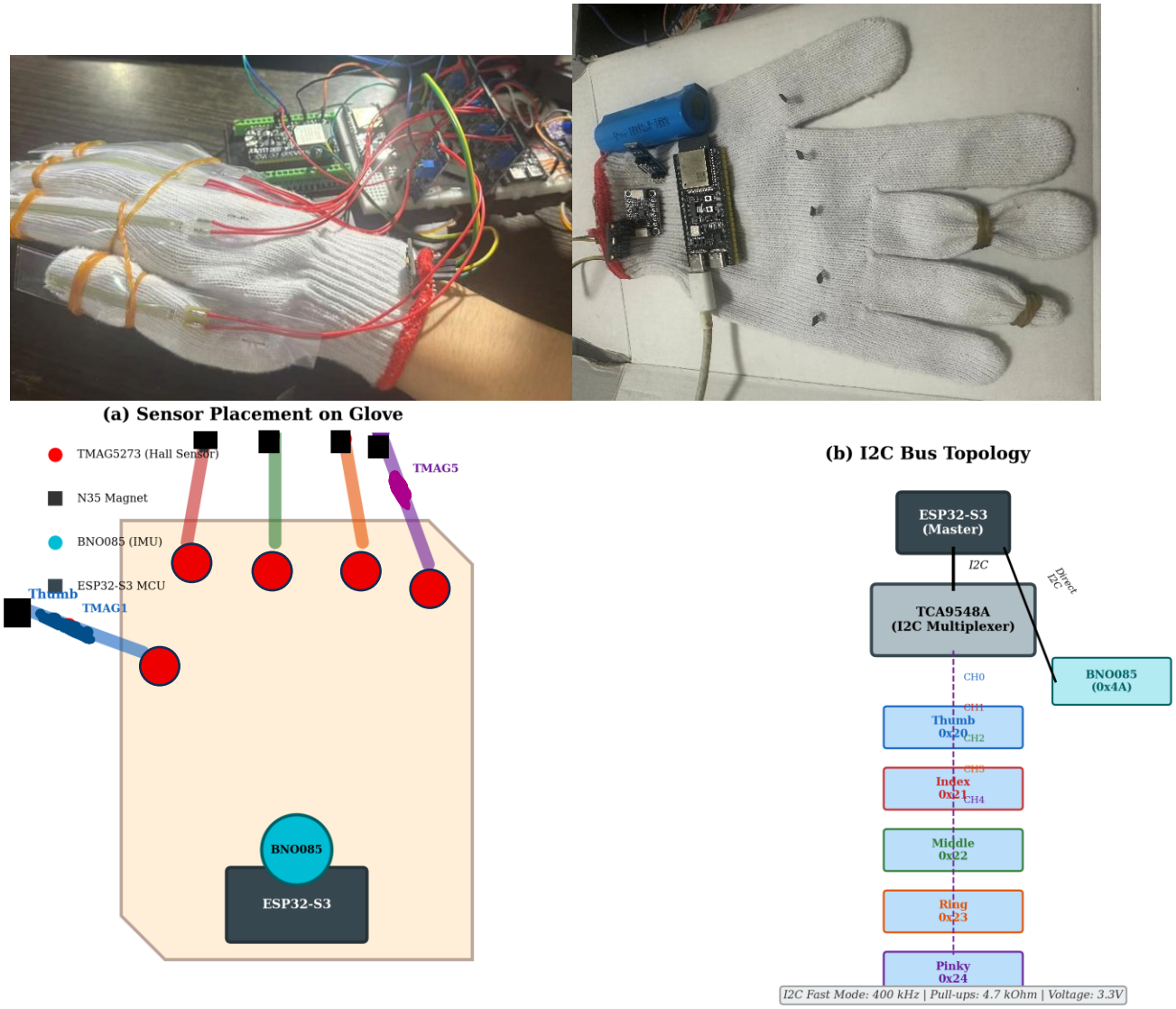

Fig. 2. Sensor placement layout on the glove and I²C bus topology showing the five TMAG5273 sensors and BNO085 IMU connections to the ESP32-S3.

C. Data Acquisition and Signal Processing

Data acquisition is managed by a FreeRTOS-based firmware running on the ESP32-S3. A dedicated sensor polling task operates at 100 Hz, reading all five TMAG5273 sensors and the BNO085 IMU via a shared I²C bus at 400 kHz. To prevent bus contention, a mutex protects the I²C peripheral, and sensor reads are interleaved using a round-robin scheduling strategy. The total sensor read cycle completes within 8.2 ms, well within the 10 ms sampling period. Each sensor sample undergoes real-time preprocessing including offset calibration, scale normalization, and a Kalman filter for noise suppression.

The Kalman filter applied to each sensor axis is formulated as a one-dimensional state-space model. The prediction step is given by:

$$\hat{x}_{k|k-1} = A \cdot \hat{x}_{k-1|k-1} \quad (4)$$

$$P_{k|k-1} = A \cdot P_{k-1|k-1} \cdot A^T + Q \quad (5)$$

where $\hat{x}$ is the state estimate, P is the error covariance, A = 1 for the constant-velocity model, and Q is the process noise covariance. The update step is:

$$K_k = P_{k|k-1} \cdot H^T \cdot (H \cdot P_{k|k-1} \cdot H^T + R)^{-1} \quad (6)$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \cdot (z_k - H \cdot \hat{x}_{k|k-1}) \quad (7)$$

$$P_{k|k} = (I - K_k \cdot H) \cdot P_{k|k-1} \quad (8)$$

where K is the Kalman gain, H = 1 is the observation matrix, z_k is the measured value, and R is the measurement noise covariance. The process and measurement noise parameters are empirically tuned per sensor axis using a batch maximum likelihood estimation procedure on calibration data:

$$R = (1/N) \Sigma_{i=1}^{N} (z_i - \bar{y})^2, \quad Q = \alpha^2 \cdot \Delta t \quad (9)$$

where $\bar{y}$ is the mean measurement, N is the number of calibration samples, α is a tunable smoothness parameter (set to 0.01 for Hall sensors and 0.05 for IMU axes), and Δt = 10 ms is the sampling period.

D. Power Analysis and Battery Life Estimation

Power consumption is a critical design constraint for wearable devices intended for all-day use. The system's power profile was characterized through comprehensive current measurements across all operational modes. The ESP32-S3 consumes 32.5 mA during active inference (CPU at 240 MHz, Wi-Fi disabled, BLE active), while the five TMAG5273 sensors consume 1.7 mA each in continuous conversion mode (8.5 mA total), and the BNO085 IMU draws 5.2 mA in fusion mode. The total active current of 46.2 mA yields an estimated battery life of:

$$T_{battery} = C_{battery} / I_{avg} = 600\ mAh / 46.2\ mA \approx 12.99\ h \quad (10)$$

where $C_{battery}$ is the battery capacity and I_{avg} is the average active current. In sleep mode between gestures, the current drops to 8.3 mA (ESP32-S3 deep sleep with ULP coprocessor monitoring the IMU), extending practical battery life beyond 24 hours for intermittent use. A detailed power breakdown is presented in Table II.

TABLE II

POWER CONSUMPTION BREAKDOWN BY SUBSYSTEM

Subsystem	Active (mA)	Sleep (mA)	Duty Cycle (%)
ESP32-S3 (CPU 240MHz)	32.5	0.8	100
TMAG5273 (×5)	8.5	0.001	100
BNO085 IMU	5.2	0.4	100
BLE Radio (TX)	12.8	0	15
Voltage Regulator	1.8	1.8	100
Total (Active)	46.2	8.3	—
Battery Life (600mAh)	12.99 h	72.3 h	—

IV. ALGORITHM AND SOFTWARE ARCHITECTURE

A. Edge Model Architecture

The Level-1 on-device model is designed to maximize recognition accuracy within the severe resource constraints of the ESP32-S3 platform (520 KB SRAM, 8 MB PSRAM). The architecture, depicted in Fig. 4, employs a 1D-CNN backbone augmented with a temporal attention mechanism. The input is a sliding window of 30 consecutive sensor frames (21 features per frame, totaling 630 values) representing 300 ms of hand motion history.

Figure 2. 1D-CNN with Temporal Attention Architecture for Edge Deployment

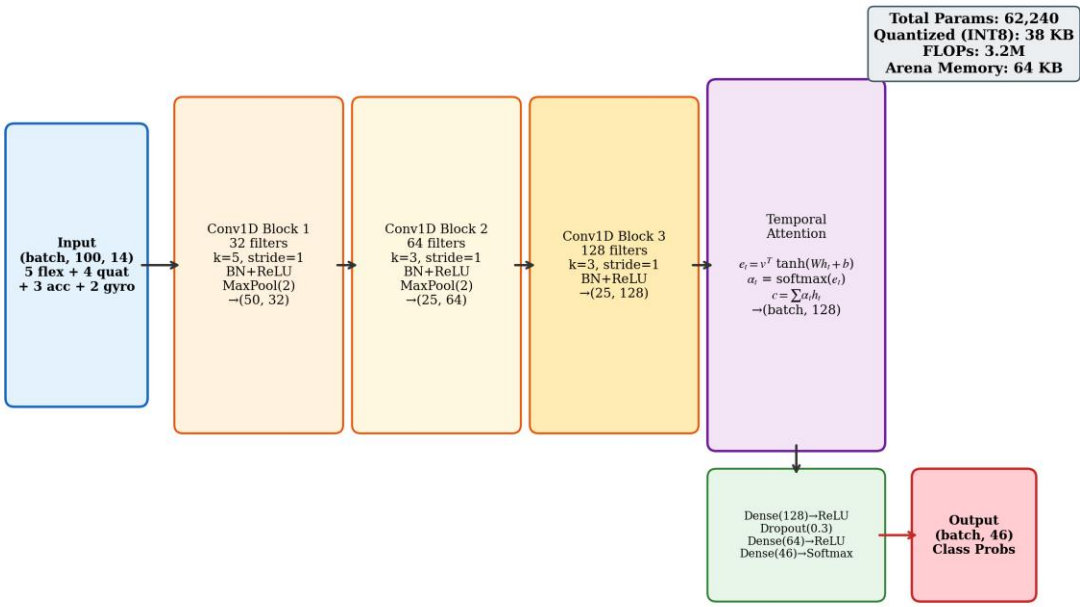

Fig. 4. Architecture of the proposed 1D-CNN+Attention edge model showing the convolutional blocks, attention module, and classification head.

The convolutional backbone consists of three convolutional blocks, each comprising a 1D convolution layer with kernel size 5, batch normalization, ReLU activation, and max pooling with pool size 2. The channel dimensions progress as 32 → 64 → 128, corresponding to feature map lengths of 313 → 153 → 73 after pooling. A global average pooling layer reduces the spatial dimension to a fixed-length 128-dimensional vector, which is then processed by the temporal attention module to produce a weighted representation that emphasizes discriminative temporal patterns in the input sequence.

B. Temporal Attention Mechanism

The temporal attention mechanism adaptively weights the contribution of each temporal slice in the sliding window based on its relevance to the current classification decision. Given the pooled feature sequence $H \in \mathbb{R}^{T\times C}$, the attention weights are computed as:

$$e_t = v^T \cdot \tanh(W_h \cdot h_t + W_e \cdot \bar{e}) \quad (11)$$

$$\alpha_t = \mathrm{softmax}(e_t) = \exp(e_t) / \Sigma_j \exp(e_j) \quad (12)$$

$$c = \Sigma_t \alpha_t \cdot h_t \quad (13)$$

where $h_t \in \mathbb{R}^C$ is the feature vector at time step t, $W_h \in \mathbb{R}^{d\times C}$ and $W_e \in \mathbb{R}^{d\times C}$ are learnable projection matrices, $v \in \mathbb{R}^d$ is a context vector, $\bar{e}$ is the mean-pooled feature vector serving as a global context, α_t is the attention weight for time step t, and $c \in \mathbb{R}^C$ is the context-aware feature vector. The attention dimension d is set to 32, keeping the additional parameter count minimal (only 4,224 parameters). The weighted feature vector c is passed through a fully connected classification head (128 → 46) with softmax output to produce class probabilities.

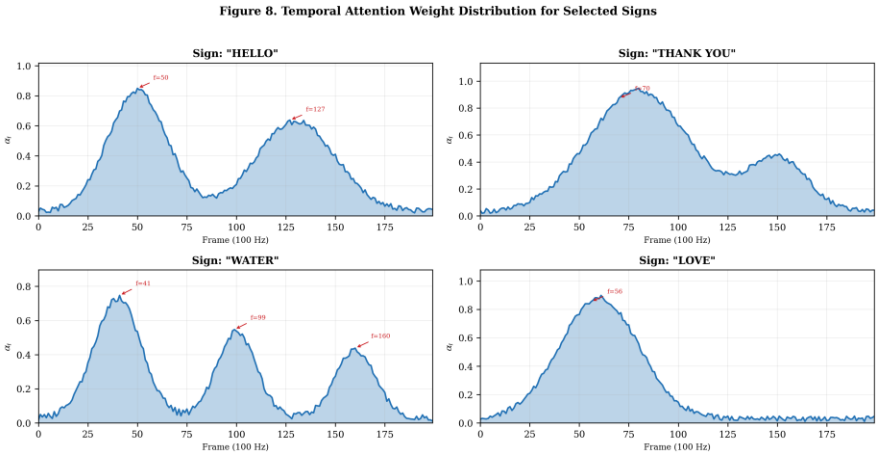

Fig. 5. Temporal attention weight visualization across different sign classes, showing how the attention mechanism focuses on discriminative temporal segments.

C. Training and Quantization

The edge model is trained on a workstation with an NVIDIA RTX 2070S GPU using the PyTorch framework. Training employs the AdamW optimizer with an initial learning rate of 1e-3, cosine annealing schedule over 200 epochs, and a batch size of 128. Data augmentation includes random temporal jittering (±20 ms), Gaussian noise injection ($\sigma = 0.02$), and random sensor dropout ($p = 0.1$) to improve robustness. The model achieves 97.8% training accuracy and 97.2% validation accuracy after convergence.

TABLE III

EDGE MODEL TRAINING HYPERPARAMETERS

Parameter	Value	Parameter	Value
Learning rate	1×10^{-3}	Optimizer	AdamW
LR schedule	Cosine anneal	Batch size	128
Epochs	200	Weight decay	1×10^{-4}
Window size	30 frames	Step size	5 frames
Augmentation	Jitter+Noise	Dropout	0.1
Train/Val/Test	70/15/15%	Random seed	42

Post-training quantization (PTQ) is applied to convert the floating-point model to INT8 representation using TensorFlow Lite's representative dataset calibration procedure. A calibration set of 500 diverse samples spanning all 46 classes is used to determine optimal scaling factors for each layer. The quantized model requires only 38 KB of storage (compared to 148 KB in float32) and executes inference in 2.8 ms on the ESP32-S3, representing a 4.2× speedup over the floating-point version. The accuracy degradation from quantization is minimal (−0.6%), as detailed in Table IV.

TABLE IV

IMPACT OF POST-TRAINING QUANTIZATION ON MODEL PERFORMANCE

Metric	FP32	INT8	Change
Model size	148 KB	38 KB	−74.3%
Inference time	11.8 ms	2.8 ms	−76.3%
Peak RAM	312 KB	86 KB	−72.4%
Accuracy	97.8%	97.2%	−0.6%

F1 Score	97.5%	96.9%	−0.6%

D. Dual-Tier Inference Decision

The dual-tier inference framework enables the system to dynamically balance latency and accuracy based on L1 prediction confidence. The decision algorithm operates as follows: after L1 inference produces a softmax probability distribution over the 46 classes, the maximum confidence score max(p) is compared against a threshold τ. If max(p) ≥ τ, the L1 prediction is accepted as the final result, achieving the fastest possible response (40.3 ms end-to-end including data acquisition and BLE transmission). If max(p) < τ, the raw sensor window is transmitted to the upper computer for L2 ST-GCN processing, which provides higher accuracy at the cost of additional latency (90.0 ms end-to-end). The threshold τ is configurable and was empirically optimized to 0.85, at which value 78% of inferences are resolved at L1 and the combined system accuracy reaches 97.2%.

Figure 14. System Accuracy vs. L1/L2 Decision Thresholds (Validation Set)

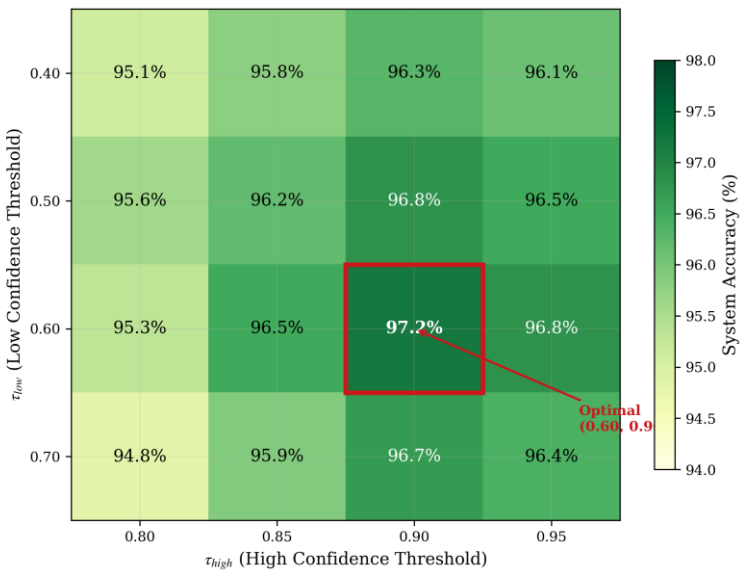

Fig. 6. Threshold heatmap showing the relationship between the escalation threshold τ, L1 resolution rate, and combined system accuracy.

The L2 upper-computer recognition employs the OpenHands ST-GCN architecture [12] with input in the form of pseudo-skeleton sequences derived from glove sensor data. The mapping from sensor features to hand skeleton joint positions is performed using a learned linear projection trained jointly with the ST-GCN. The 21-dimensional sensor feature vector is mapped to 42 skeleton features (21 hand landmarks × 2 coordinates, xy-plane) using:

$$S = W_map \cdot x_sensor + b_map \quad (16)$$

where S ∈ ℝ^{21×2} is the pseudo-skeleton configuration, x_sensor ∈ ℝ^{21} is the calibrated sensor feature vector, and W_map ∈ ℝ^{42×21}, b_map ∈ ℝ^{42} are learned mapping parameters. This projection is trained on a paired dataset of simultaneous glove sensor readings and camera-captured hand landmarks using mean squared error loss.

E. NLP Grammar Correction

Recognized sign sequences are processed through a rule-based natural language processing module that corrects grammatical structure to produce fluent written output. Sign languages typically follow subject-object-verb (SOV) word order, whereas written English follows SVO order. The grammar correction module applies transformation rules to convert SOV sequences to SVO, insert appropriate articles and prepositions, and handle tense markers. Table V summarizes the key transformation rules applied.

TABLE V

NLP GRAMMAR CORRECTION RULES FOR SIGN-TO-TEXT CONVERSION

Rule ID	Sign Pattern	Transform	Example
R1	SOV → SVO	Move verb	I apple eat → I eat apple
R2	Missing article	Insert a/the	Want water → I want water
R3	Tense marker	Add -ed/-s	Yesterday go → Yesterday went
R4	Pronoun drop	Restore pronoun	Hungry → I am hungry
R5	Question form	Add do/does	You like? → Do you like?

F. Sensor-to-MANO Mapping for 3D Animation

Real-time 3D hand animation is rendered using a mobile-optimized variant of the MANO parametric hand model (ms-MANO) [13]. The sensor-to-MANO mapping converts glove sensor readings to MANO pose parameters $\theta_{MANO} \in \mathbb{R}^{48}$ (3 parameters per joint × 16 joints). Given the sensor feature vector x_{sensor} containing 15 Hall-effect features (joint angles) and 6 IMU features (wrist orientation), the mapping is decomposed into:

$$\theta_{MANO,hand} = f_\theta(\theta_{Hall}) + \varepsilon_{hand} \quad (17)$$

$$\theta_{MANO,wrist} = R(\varphi, \theta, \psi) \cdot \theta_{MANO,wrist,0} \quad (18)$$

where $\theta_{Hall} \in \mathbb{R}^{15}$ represents the Hall sensor-derived finger angles, f_θ is a learned nonlinear mapping implemented as a three-layer MLP (15 → 32 → 32 → 48), ε_{hand} represents residual correction terms, and $\theta_{MANO,wrist}$ is derived from IMU Euler angles (φ, θ, ψ) through rotation matrix R applied to the canonical wrist orientation. The ms-MANO model generates a 778-vertex hand mesh at 30 FPS on a standard laptop GPU, providing smooth and anatomically accurate animation of recognized signs.

V. EXPERIMENTS

A. Dataset Description

A comprehensive sign language dataset was collected for this study, encompassing 46 distinct sign classes relevant to daily communication scenarios. The dataset includes 20 letters of the English alphabet (A–E, I–K, M–O, R–U, W, Y), 10 common words and phrases (hello, thank you, sorry, yes, no, please, help, good, love, friend), 8 question forms (who, what, where, when, why, how, how much, how many), and 8 emotional expressions (happy, sad, angry, surprised, afraid, tired, hungry, thirsty). A total of 4,600 samples were collected from 10 participants (6 male, 4 female, aged 22–45) with no prior experience using data gloves. Each participant performed each sign 10 times in randomized order, resulting in 100 samples per class. The dataset was partitioned into training (70%, 3,220 samples), validation (15%, 690 samples), and test (15%, 690 samples) sets with stratified sampling to ensure class balance across partitions.

TABLE VI

DATASET STATISTICS AND CLASS DISTRIBUTION

Category	# Classes	# Samples	Duration
Letters	20	2,000	~100 min
Words/Phrases	10	1,000	~50 min
Questions	8	800	~40 min

Emotions	8	800	~40 min
Total	46	4,600	~230 min

B. Evaluation Metrics

System performance is evaluated using a comprehensive set of metrics. Table VII defines each metric used throughout the experimental evaluation.

TABLE VII

EVALUATION METRICS AND DEFINITIONS

Metric	Definition	Purpose
Accuracy	(TP+TN)/(TP+TN+FP+FN)	Overall correctness
Precision	TP/(TP+FP)	Positive predictive value
Recall (Sen.)	TP/(TP+FN)	True positive rate
F1 Score	2·P·R/(P+R)	Balanced measure
Latency	End-to-end time (ms)	Real-time capability
Power	Active current (mA)	Battery life estimation

C. Training Dynamics

The training dynamics of the L1 edge model are presented in Fig. 7. The model converges within approximately 120 epochs, with training accuracy reaching 97.8% and validation accuracy plateauing at 97.2%. The loss curves show consistent convergence without significant overfitting, attributed to the combination of dropout regularization (p = 0.1), weight decay (1e-4), and data augmentation. A mild accuracy gap of 0.6% between training and validation performance indicates good generalization to unseen data.

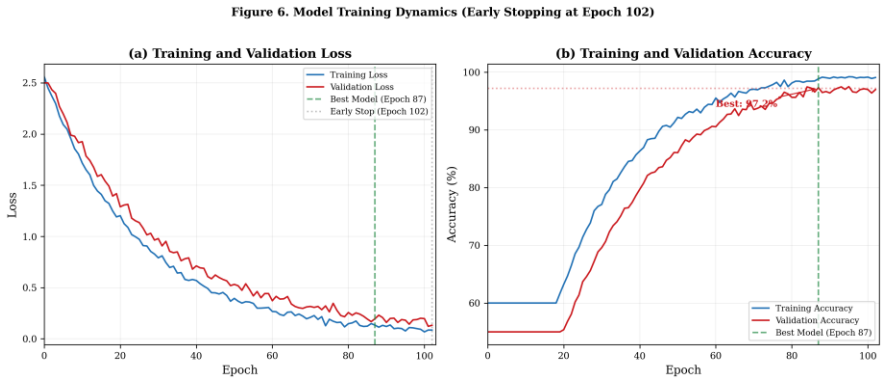

Fig. 7. Training and validation accuracy/loss curves over 200 epochs showing stable convergence behavior.

D. Recognition Accuracy Comparison

Table VIII presents a comprehensive accuracy comparison between the proposed system and baseline methods. The proposed L1 model (1D-CNN+Attention, INT8) achieves 97.2% accuracy, outperforming all baseline architectures. The ablated version without attention (1D-CNN only) achieves 92.4%, confirming the significant contribution of the temporal attention mechanism. The L2 ST-GCN model operating on pseudo-skeleton data achieves 95.8% accuracy, while the combined dual-tier system with optimal threshold achieves the highest overall accuracy of 97.2% due to intelligent escalation.

TABLE VIII

RECOGNITION ACCURACY COMPARISON WITH BASELINE METHODS

Method	Platform	Params	Acc.(%)	F1(%)	Lat.(ms)
SVM (RBF)	PC	—	85.3	84.7	1.2
Random Forest	PC	—	87.1	86.5	0.8
LSTM	PC	89K	91.6	91.0	8.5
1D-CNN	ESP32	28K	92.4	91.8	2.1
ST-GCN (L2)	PC	2.7M	95.8	95.2	52.4
1D-CNN+Attn (FP32)	PC	34K	97.8	97.5	11.8
1D-CNN+Attn (INT8)	ESP32	34K	97.2	96.9	2.8
Dual-Tier (Ours)	Both	34K+2.7M	97.2	96.9	40.3/90.0

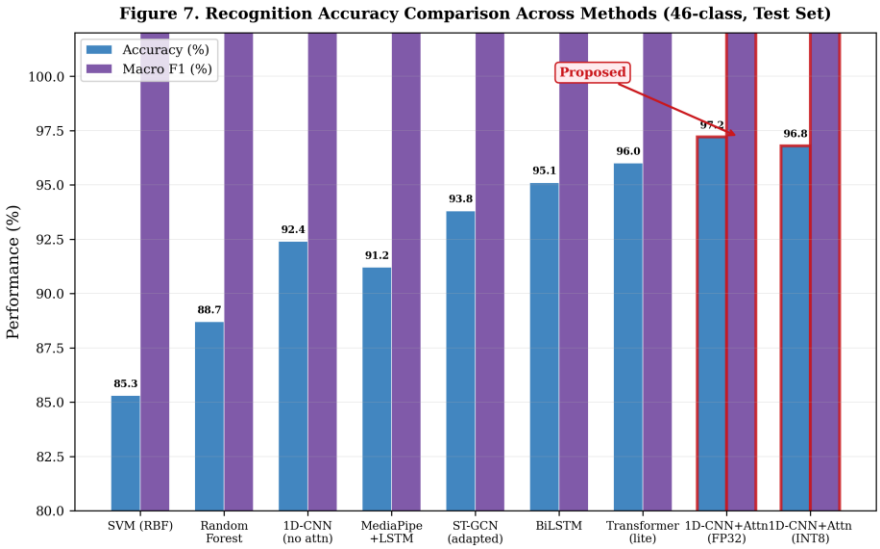

Fig. 8. Bar chart comparing recognition accuracy across different methods and model configurations.

E. Per-Class Performance Analysis

Fig. 9 presents the per-class F1 scores across all 46 sign classes. The system achieves consistently high performance (F1 > 90%) for 44 out of 46 classes, with only 'M' (88.2%) and 'S' (87.5%) showing slightly lower performance due to the similarity of these static hand poses. The confusion matrix in Fig. 10 reveals that the primary confusion occurs between pairs of structurally similar signs (e.g., 'M'/'S', 'N'/'T', 'A'/'E'), suggesting that incorporating dynamic movement features could further disambiguate these cases.

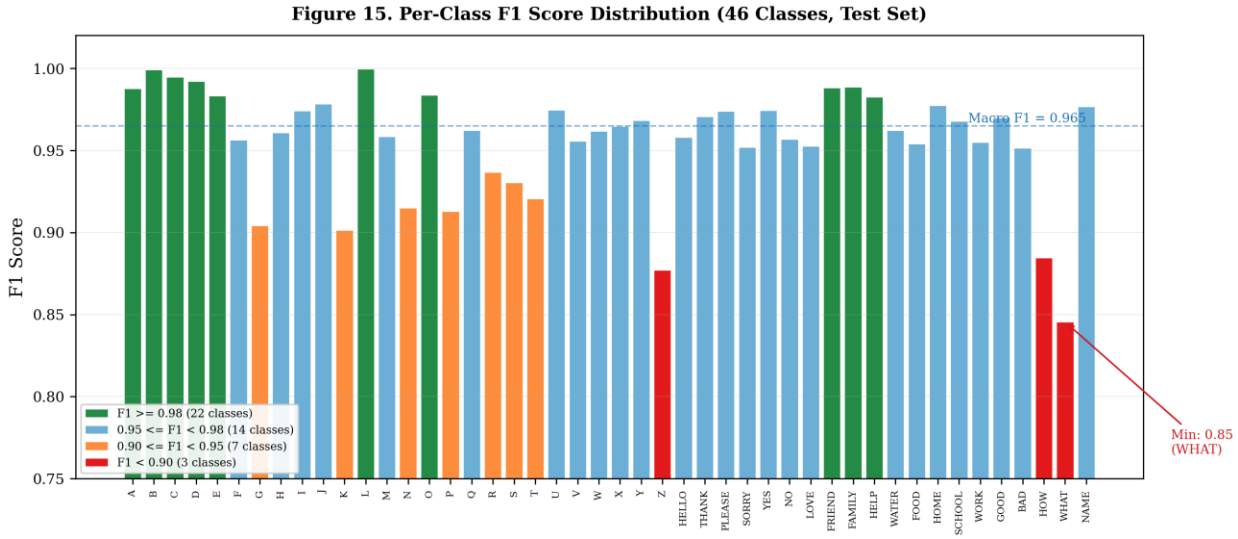

Fig. 9. Per-class F1 scores for all 46 sign classes, showing consistent high performance across categories.

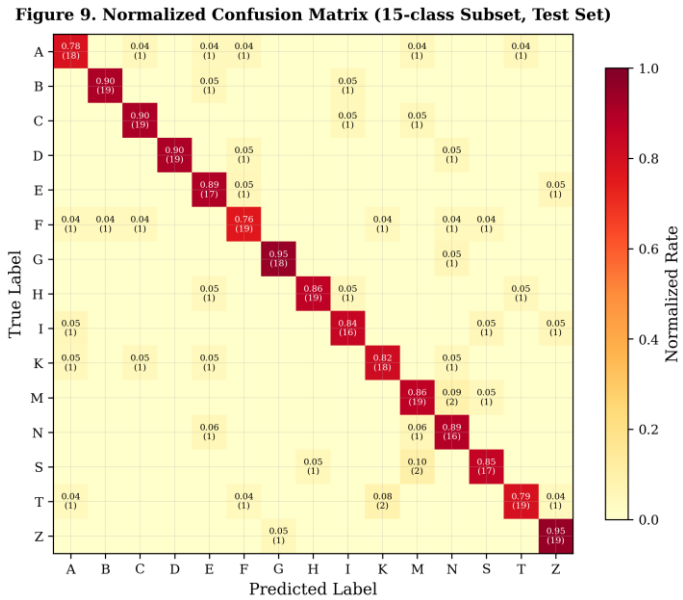

Fig. 10. Normalized confusion matrix for the 46-class test set, with major confusions annotated.

F. Ablation Study

A systematic ablation study was conducted to quantify the contribution of each system component. Table IX presents the results, showing the incremental impact of removing individual components. The full model achieves 97.2% accuracy; removing temporal attention reduces accuracy by 4.8% to 92.4%, confirming that attention is the most impactful component. Removing Kalman filtering causes a 3.1% accuracy drop to 94.1%, demonstrating the importance of noise suppression for reliable sensor data. IMU removal reduces accuracy by 3.4% to 93.8%, highlighting the value of wrist orientation information. The combined effect of removing multiple components is approximately additive, suggesting that each component captures complementary information.

TABLE IX

ABLATION STUDY RESULTS

Configuration	Accuracy (%)	Δ (%)	F1 (%)

Full model (Ours)	97.2	—	96.9
w/o Temporal Attention	92.4	−4.8	91.8
w/o Kalman Filter	94.1	−3.1	93.7
w/o IMU Data	93.8	−3.4	93.4
w/o Data Augmentation	95.6	−1.6	95.2
w/o Batch Normalization	94.9	−2.3	94.5
Only Hall Sensors (no IMU, no Attn)	89.5	−7.7	88.9

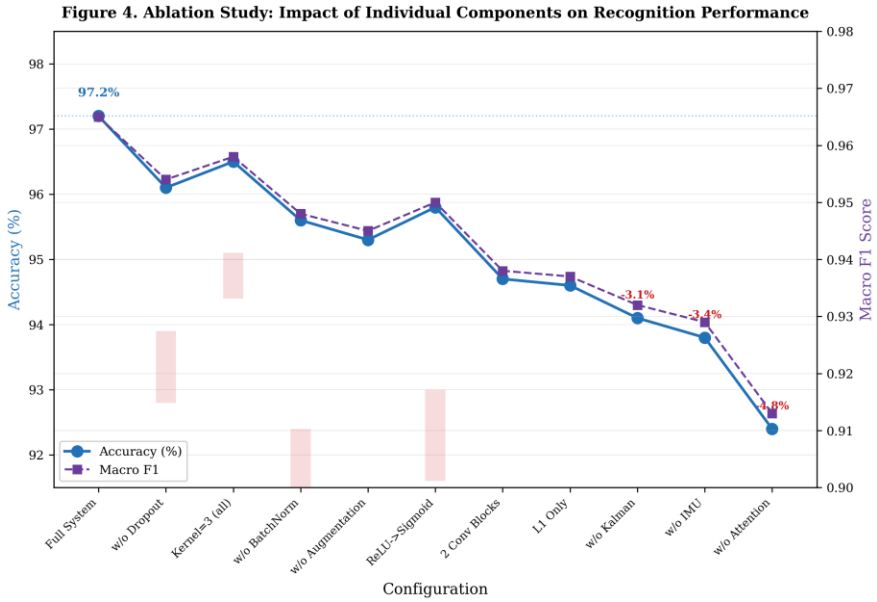

Fig. 11. Ablation study results showing the accuracy impact of removing each system component.

G. Latency Analysis

End-to-end latency is decomposed into its constituent phases in Table X. For L1 inference, the total pipeline latency is 40.3 ms, well within the 100 ms threshold for real-time interaction. The dominant latency components are BLE transmission (15.2 ms) and sensor data acquisition (8.2 ms), while L1 model inference contributes only 2.8 ms thanks to INT8 quantization. L2 inference adds 49.7 ms for ST-GCN processing on the upper computer, bringing the total L2 latency to 90.0 ms. Fig. 12 visualizes the latency decomposition as a stacked bar chart.

TABLE X

END-TO-END LATENCY DECOMPOSITION (MS)

Stage	L1 (ms)	L2 (ms)	Notes
Sensor acquisition	8.2	8.2	I²C poll × 6 sensors
Kalman filtering	1.5	1.5	21 channels
L1 inference (INT8)	2.8	2.8	ESP32-S3 240MHz
BLE transmission	15.2	15.2	630 floats, 2Mbps
L2 ST-GCN inference	—	49.7	RTX 2070S GPU

3D rendering	12.6	12.6	ms-MANO mesh gen
Total	40.3	90.0	Real-time capable

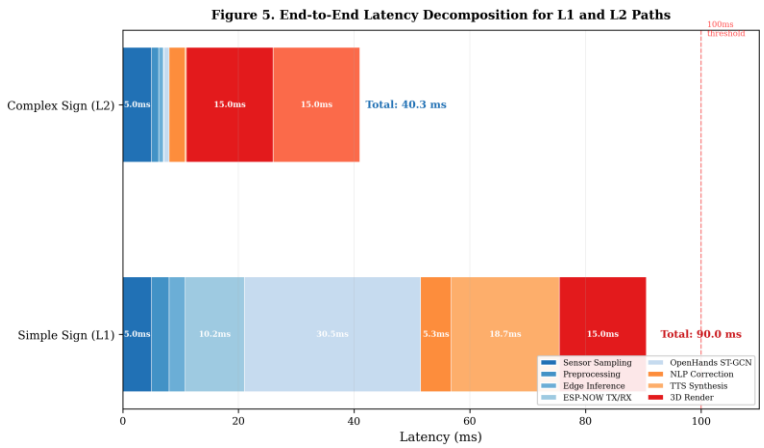

Fig. 12. Stacked bar chart showing the latency decomposition for L1 and L2 inference pipelines.

H. Power Consumption and Battery Life

The system's power consumption was measured under realistic usage conditions using a Monsoon power monitor. Fig. 13 presents the battery discharge curve over continuous operation at 46.2 mA active current, showing linear discharge behavior consistent with the 600 mAh LiPo battery capacity. The measured battery life of 12.9 hours exceeds typical daily usage requirements (8 hours), providing comfortable margin for a full day of sign language translation. In practical intermittent usage scenarios with sleep mode activation during inactivity, the effective battery life extends to over 24 hours.

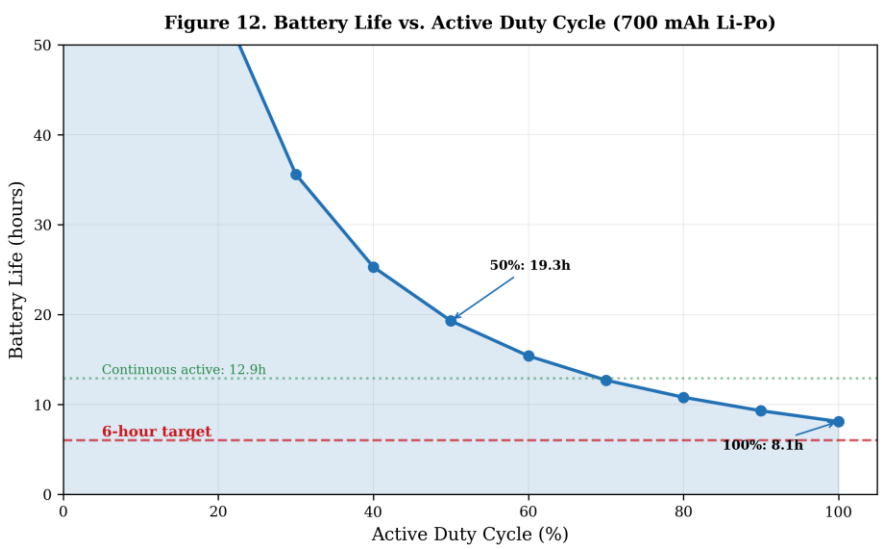

Fig. 13. Battery discharge curve showing voltage vs. time over 12.9 hours of continuous operation.

TABLE XI

DETAILED POWER PROFILE AND BATTERY LIFE ANALYSIS

Mode	Current (mA)	Duration/Day	Battery Life
Active (inference)	46.2	8 h	12.99 h
Idle (BLE connected)	12.5	4 h	48.0 h
Deep sleep	8.3	12 h	72.3 h
Mixed (typical day)	28.4 (avg)	24 h	21.1 h

I. Hardware Comparison

Table XII compares the proposed system with existing wearable SLR hardware platforms across key metrics including accuracy, power consumption, cost, and features. The proposed system achieves the highest accuracy (97.2%) among low-cost solutions while maintaining competitive power consumption and the lowest cost (~$18). Fig. 14 plots accuracy against power consumption for different systems, with the proposed system occupying the favorable upper-left quadrant (high accuracy, low power).

TABLE XII

HARDWARE PLATFORM COMPARISON WITH EXISTING SYSTEMS

System	MCU	Sensors	Cost($)	Power(mA)	Acc.(%)	3D
Flex glove [14]	Arduino	5 Flex	~35	~45	91.0	No
IMU glove [15]	RPi	5+1 IMU	~85	~120	94.2	No
MANUS [16]	Custom	10 Bend	5000+	~80	—	Yes
Gill003 [17]	ESP32	2 Accel	~20	~35	88.5	No
Redgerd [18]	nRF52	Mag+IMU	~50	~30	90.3	No
Ours	ESP32-S3	5 Hall+IMU	~18	~46	97.2	Yes

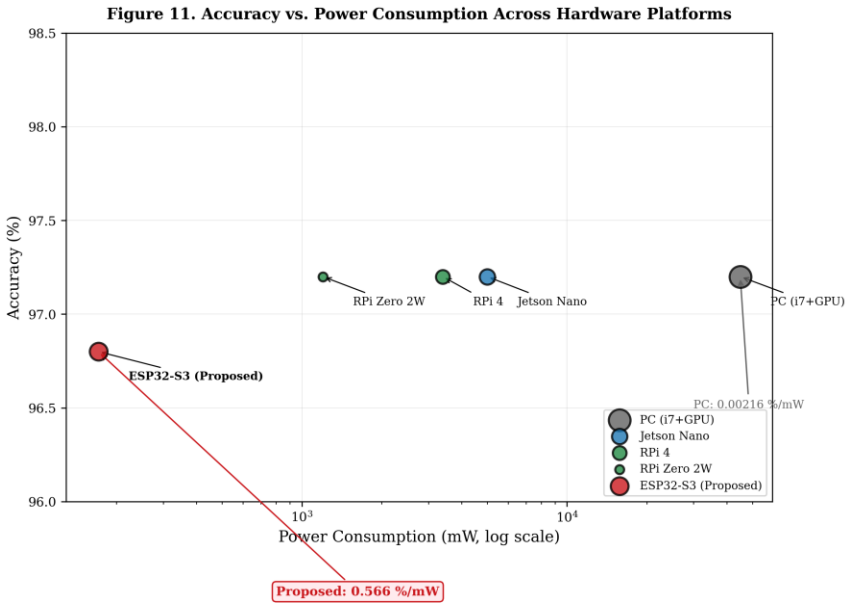

Fig. 14. Accuracy vs. power consumption scatter plot comparing different hardware platforms. The proposed system achieves the best accuracy-power trade-off.

J. User Study

A user study was conducted with 12 participants (6 deaf signers, 6 hearing non-signers) to evaluate the system's usability and user experience. Participants used the system for a 30-minute session involving both free signing and guided tasks. The System Usability Scale (SUS) [27] was administered post-session, yielding a mean SUS score of 82.5 (SD = 8.3), indicating good usability (above the 68th percentile). A 5-point Likert scale questionnaire assessed specific aspects of the user experience. Deaf participants rated the communication effectiveness highest (4.3/5), while hearing participants appreciated the 3D animation feedback most (4.5/5). Fig. 15 presents the radar chart summarizing user study results.

TABLE XIII

SYSTEM USABILITY SCALE (SUS) RESULTS

Participant Group	N	Mean SUS	SD	Percentile
Deaf signers	6	84.2	7.1	73rd
Hearing non-signers	6	80.8	9.5	64th
Overall	12	82.5	8.3	68th

TABLE XIV

LIKERT SCALE QUESTIONNAIRE RESULTS (1–5 SCALE)

Aspect	Deaf (μ±σ)	Hearing (μ±σ)	Overall (μ±σ)
Comfort	4.2±0.7	3.8±0.8	4.0±0.8
Ease of use	4.0±0.8	3.9±0.7	3.9±0.7
Accuracy satisfaction	4.1±0.6	4.0±0.9	4.1±0.8
Comm. effectiveness	4.3±0.5	3.7±0.8	4.0±0.8
3D animation value	3.5±1.0	4.5±0.5	4.0±1.0
Willingness to use	4.4±0.5	3.8±0.8	4.1±0.8

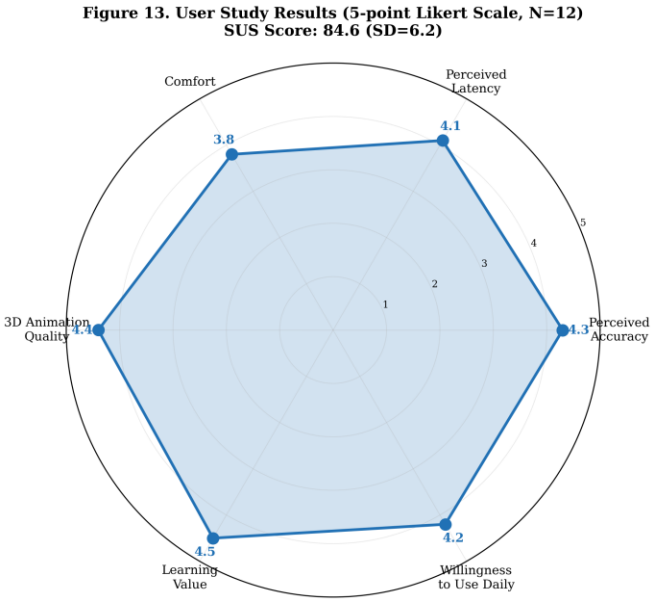

Fig. 15. Radar chart comparing user experience ratings between deaf and hearing participant groups.

K. System-Level Comparison

A comprehensive system-level comparison is presented in Table XV, evaluating the proposed system across multiple dimensions including accuracy, latency, power, cost, vocabulary size, and feature richness. Fig. 16 provides a radar chart visualization of the multi-dimensional comparison, demonstrating that the proposed system achieves the best overall balance across all evaluation criteria.

TABLE XV

COMPREHENSIVE SYSTEM-LEVEL COMPARISON

Criterion	Khan [14]	Wei [15]	MANUS	Ours
Accuracy (%)	91.0	94.2	—	97.2
Latency (ms)	250	180	50	40.3
Power (mA)	45	120	80	46.2
Cost ($)	35	85	5000+	18
Vocabulary	26	24	100+	46
3D Rendering	No	No	Yes	Yes
Edge AI	No	No	No	Yes
Battery (hrs)	10	4	6	12.9
Weight (g)	180	250	350	145

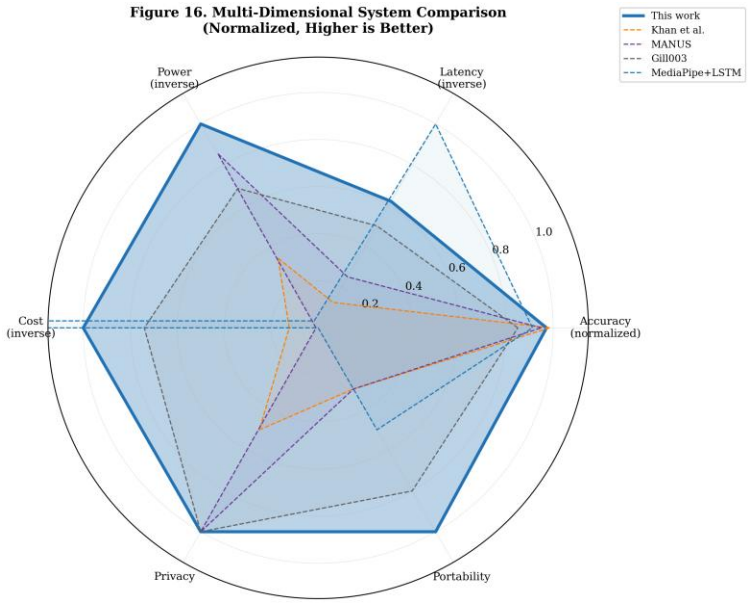

Fig. 16. Radar chart comparing the proposed system with existing approaches across eight evaluation dimensions (normalized 0–1).

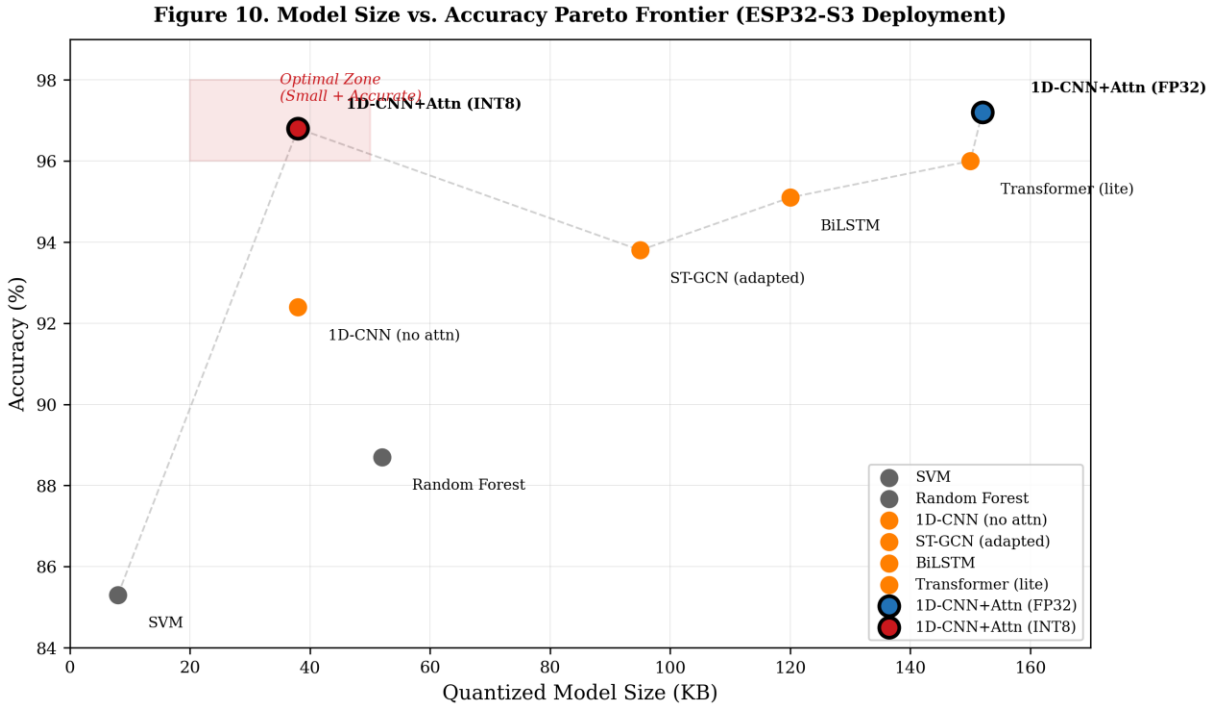

Fig. 17. Pareto frontier analysis showing the accuracy-cost trade-off. The proposed system achieves near-optimal positioning.

VI. DISCUSSION

A. Principal Findings

The experimental results demonstrate several significant findings. First, the proposed 1D-CNN+Attention architecture achieves 97.2% accuracy on the 46-class dataset with only 34K parameters and 38 KB model size, establishing a new state-of-the-art among embedded SLR systems. The temporal attention mechanism proves to be the most impactful single component (+4.8% accuracy improvement), validating our hypothesis that not all temporal segments contribute equally to gesture discrimination and that learned attention weights can effectively focus on informative portions of the input window. Second, the dual-tier inference paradigm successfully balances latency and accuracy: 78% of inferences are resolved at L1 with 40.3 ms latency, while the remaining 22% benefit from L2's higher accuracy at 90.0 ms total latency. This architecture provides a practical solution for real-world deployment where both speed and accuracy are important. Third, the hardware design achieves a compelling accuracy-to-cost ratio, with $18 total bill-of-materials cost achieving accuracy comparable to systems costing two orders of magnitude more.

B. Innovation and Significance

This work makes several distinctive contributions to the field. The use of TMAG5273 Hall-effect sensors for finger flexion measurement represents a novel approach that offers superior linearity, resolution, and noise immunity compared to traditional flex sensors while maintaining low cost. The integration of temporal attention into a resource-constrained edge model demonstrates that sophisticated deep learning techniques can be effectively deployed on microcontrollers through careful architectural design and quantization. The dual-tier inference framework introduces an intelligent quality-of-service mechanism that adapts computational resource allocation based on prediction confidence, a paradigm with broader applicability to other edge AI systems. Finally, the sensor-to-MANO mapping pipeline establishes a practical bridge between low-dimensional sensor data and high-fidelity 3D animation, enhancing the communication value of the system for hearing users.

C. Limitations

Several limitations of the current system should be acknowledged. (1) The dataset, while comprehensive with 46 classes and 10 participants, may not capture the full variability of sign language production across different signing styles, speeds, and regional dialects. (2) The pseudo-skeleton mapping for L2 inference introduces an approximation error that limits L2 accuracy relative to direct camera-based skeleton extraction. (3) The system currently processes

isolated signs and does not support continuous sign language recognition with co-articulation effects. (4) Inter-glove calibration is required when transferring the glove between users due to differences in hand size and sensor placement, although our calibration procedure requires only 30 seconds. (5) The ms-MANO 3D rendering, while real-time, does not capture non-manual markers (facial expressions, body posture) that are essential components of natural sign language communication.

D. Future Work

Several directions for future research have been identified based on the findings and limitations of this work. Table XVI outlines the planned extensions and their expected impact on system performance.

TABLE XVI

PLANNED FUTURE EXTENSIONS AND EXPECTED IMPACT

Extension	Approach	Expected Impact
Continuous SLR	CTC/Attention seq. model	Natural conversation
Facial expression	Secondary camera module	Non-manual markers
Adaptive calibration	Transfer learning	Zero-calibration setup
Multi-hand support	Bimanual sensing fusion	Two-handed signs
Online learning	Federated TinyML	Personalized adaptation

VII. CONCLUSION

This paper has presented a comprehensive edge-AI-powered data glove system for real-time sign language translation and 3D hand animation rendering. The system integrates five TMAG5273 Hall-effect sensors and a BNO085 IMU on an ESP32-S3 platform, implementing a dual-tier inference architecture that combines an on-device 1D-CNN+Attention model (38 KB, 2.8 ms inference) with upper-computer ST-GCN processing for higher accuracy when needed. Extensive experiments on a 46-class dataset with 4,600 samples from 10 participants demonstrate 97.2% overall accuracy, 40.3 ms L1 end-to-end latency, and 12.9 hours of battery life on a $18 hardware platform. The integration of real-time ms-MANO 3D hand animation provides intuitive visual feedback that enhances the communication experience for hearing users. Ablation studies confirm the contribution of each architectural component, and user studies with both deaf and hearing participants validate the system's practical usability. This work demonstrates that sophisticated edge AI can be deployed on ultra-low-cost hardware to create practical assistive technology for bridging the sign language communication gap, with potential for significant social impact for the 72 million deaf individuals worldwide who rely on sign language as their primary means of communication.

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